

A Novel Pair Bound for Kuramoto-Type Stability: Machine-Checked Proofs via an Algebraic Lyapunov Technique

Abstract

We introduce two novel proof techniques for Kuramoto-type stability problems and present the first machine-checked formalization of any Kuramoto model result. The complex OA and full PDE results are conditional on two cited axioms (nonlinear Landau damping and OA manifold attractivity).

Contribution 1: The pair bound. A pointwise algebraic pair inequality, lifted to the continuum double integral via Fubini, that proves the L^2 Lyapunov derivative $\dot{V} \leq 0$ for the scalar dissipative mean-field equation. This is a purely algebraic technique with no precedent in the Kuramoto literature, replacing spectral theory, Volterra equations, and Fourier analysis approaches.

Contribution 2: Rotation cancellation. For the standard complex OA equation $\dot{z} = -i\omega z + (K/2)(\bar{\eta} - \eta z^2)$, the rotation term $-i\omega$ cancels exactly in the Lyapunov derivative after equilibrium subtraction: $\text{Re}(\bar{w} \cdot (-i\omega w)) = 0$ for $w = z - z^*$, since $|w|^2 \in \mathbb{R}$. This bridges the dissipative and rotational formulations.

Contribution 3: Formalization. The Lean 4 formalization (Mathlib v4.30.0-rc1) spans **255 Lean files** (commit 58b6194); the main theorem `real_scalar_complete` has **0 sorry** and **0 project axioms**; the declaration `real_scalar_complete` itself produces no sorry warning (the imported file `KuramotoGlobal.lean` contains unrelated sorry's in other theorems that are not used by our proof chain). The main theorem `real_scalar_complete` (**0 sorry, 0 axioms**) proves: for the scalar dissipative mean-field equation (2) with $\gamma > 0$ pointwise, finite first moment $\int \gamma g < \infty$, and initial data in the basin $V(0) < (r^*)^2$, the order parameter converges $r(t) \rightarrow r^*$. Body persistence is *derived internally*, not assumed. The proof wires: pair bound ($\dot{V} \leq 0$) $\rightarrow$ Cauchy–Schwarz ($r \geq r^* - \sqrt{V(0)} > 0$) $\rightarrow$ ODE barrier (body persistence from $r > 0$) $\rightarrow$ Barbalat ($V \rightarrow 0$) $\rightarrow$ convergence ($r \rightarrow r^*$). The theorem also requires initial body coherence ($\alpha(\omega, 0) \geq \delta_0 > 0$ on $\{\gamma \leq M\}$ for each M), which holds for initial data near the PLS. When $\gamma(\omega) \lesssim 1 + |\omega|$, this includes Gaussian, compactly supported, Student- t ($\nu > 1$), and mixtures satisfying the same first-moment bound. To our knowledge, this is the first fully self-contained, machine-checked stability theorem for any Kuramoto-type model.

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1 Setup and statement

1.1 The standard complex Ott–Antonsen equation

On the OA manifold, the Kuramoto–Sakaguchi PDE reduces to the *complex* OA equation:

$$\dot{z}(\omega, t) = -i\omega z + \frac{K}{2}(\bar{\eta} - \eta z^2), \quad \eta(t) = \int_{\mathbb{R}} \overline{z(\omega, t)} g(\omega) d\omega, \quad (1)$$

where $z(\omega, t) \in \mathbb{D}$ (open unit disk) and $\eta(t) \in \mathbb{C}$ is the complex order parameter. This is the standard OA reduction [5].

For symmetric g with symmetric initial data ($z(\omega, 0) = \overline{z(-\omega, 0)}$), the symmetry is preserved for all time and $\eta(t) = r(t) \in \mathbb{R}$ (`complex_oe_symmetry_preserved`, `eta_real_for_symmetric_flow`; both 0 sorry). For the Lorentzian $g(\omega) = \gamma/(\pi(\omega^2 + \gamma^2))$, residue evaluation at the pole $\omega = -i\gamma$ gives the scalar ODE $\dot{r} = (K/2 - \gamma)r - (K/2)r^3$.

1.2 A scalar dissipative model

We also study the scalar real mean-field equation

$$\dot{\alpha}(\omega, t) = -\gamma(\omega) \alpha + \frac{K}{2} r(t)(1 - \alpha^2), \quad r(t) = \int_{\mathbb{R}} \alpha(\omega, t) g(\omega) d\omega, \quad (2)$$

where $\alpha(\omega, t) \in (0, 1)$ and $\gamma(\omega) > 0$ is a dissipation rate. This is a *separate dynamical system* from (1): the term $-\gamma(\omega)\alpha$ is dissipative, whereas $-i\omega z$ in (1) is rotational. The scalar model is designed to share the same positive real self-consistency equation as the symmetric partially locked state, but its transient dynamics differ from the standard complex OA equation.

Remark 1.1 (Scope clarification). *Equation (2) is **not** derived from the complex OA equation (1) by setting $z = \alpha \in \mathbb{R}$. The subspace $\text{Im}(z) = 0$ is not invariant under (1) (since $\dot{y} = -\omega x \neq 0$ in general). The Lyapunov argument in Sections 2–2.6 applies directly to (2). The extension to the standard complex OA (1) is given in Section 6.*

1.3 The partially locked state (PLS)

Assume there exists a unique positive equilibrium (α^*, r^*) satisfying the self-consistency equation

$$r^* = \int_{\mathbb{R}} \alpha^*(\omega) g(\omega) d\omega \quad (3)$$

with $r^* \in (0, 1)$. (For the Lorentzian $g(\omega) = \gamma/(\pi(\omega^2 + \gamma^2))$, this holds when $K > K_c := 2\gamma$.) The equilibrium profile $\alpha^*(\omega) \in (0, 1)$ satisfies the pointwise equilibrium condition

$$\gamma(\omega) \alpha^*(\omega) = \frac{K}{2} r^* (1 - (\alpha^*(\omega))^2) \quad \text{for } g\text{-a.e. } \omega. \quad (4)$$

1.4 Main theorem

Theorem 1.2 (Basin stability, kuramoto_stability). *Let $\gamma(\omega) > 0$ with $\int \gamma d\mu < \infty$ (finite first moment). Let $(\alpha^*(\omega), r^*)$ satisfy (4) and (3). Let (r, α) be an OA solution:*

- (i) $\alpha(\omega, \cdot)$ solves (2) for all ω , with $r(t) = \int \alpha(\omega, t) g(\omega) d\omega$;
- (ii) $\alpha(\omega, t) \in (0, 1)$ for all $\omega, t \geq 0$ (invariant region);
- (iii) $V(0) := \int (\alpha(\omega, 0) - \alpha^*(\omega))^2 g d\omega < (r^*)^2$ (initial closeness to PLS);
- (iv) for each $M > 0$, $\exists \delta_0 > 0$ such that $\alpha(\omega, 0) \geq \delta_0$ for all ω with $\gamma(\omega) \leq M$ (initial body coherence).

Then $r(t) \rightarrow r^*$ as $t \rightarrow +\infty$.

Remark 1.3. *No persistence hypothesis is assumed. The proof derives order parameter persistence ($r(t) \geq r_{\min} > 0$) from V antitonicity and Cauchy–Schwarz, then derives body persistence on $\{\gamma \leq M\}$ for each $M > 0$ from the ODE scalar barrier. No uniform-in- ω lower bound is assumed or derived. The body-tail split uses local body persistence combined with tail vanishing from $\int \gamma g < \infty$. The only conditions beyond the OA system definition are (iii) initial closeness and (iv) initial body coherence—both hold for any initial data in a neighborhood of the PLS.*

2 The proof chain

We present the seven-step proof of Theorem 1.2 in the order that Lean verifies it. Each step is stated as a proposition and corresponds to a named lemma in the Lean formalization.

2.1 Step 1: Leibniz rule for the Lyapunov function

Define the continuum L^2 Lyapunov function

$$V(t) := \int_{\mathbb{R}} (\alpha(\omega, t) - \alpha^*(\omega))^2 g(\omega) d\omega. \quad (5)$$

Proposition 2.1 (Leibniz, `leibniz_oa_lyapunov`). *Under the hypotheses of Theorem 1.2, V is continuous on $[0, \infty)$ and for every $t > 0$:*

$$V'(t) = \int_{\mathbb{R}} 2(\alpha(\omega, t) - \alpha^*(\omega)) \dot{\alpha}(\omega, t) g(\omega) d\omega.$$

Proof. By the chain rule, $\frac{\partial}{\partial t}(\alpha(\omega, t) - \alpha^*(\omega))^2 = 2(\alpha - \alpha^*)\dot{\alpha}$. This pointwise derivative is bounded in absolute value by the integrable dominator

$$|2(\alpha - \alpha^*)\dot{\alpha}| \leq 2|\dot{\alpha}| \leq 2(\gamma(\omega) + K),$$

since $|\alpha - \alpha^*| \leq 1$ and $|\dot{\alpha}| = |\gamma(\omega)\alpha - (K/2)r(1 - \alpha^2)| \leq \gamma(\omega) + K$ for $\alpha, r \in (0, 1)$. The dominator $2(\gamma(\omega) + K)$ is integrable since $\int \gamma g d\omega < \infty$ by hypothesis. Mathlib's `hasDerivAt_integral_of_dominated_loc` then gives the stated formula for the derivative of $t \mapsto \int (\alpha(\omega, t) - \alpha^*)^2 g d\omega$. $\square$

2.2 Step 2: Pair bound and derived monotonicity

We derive $V'(t) \leq 0$ from the pair bound. Convergence $V(t) \rightarrow 0$ is obtained later by body-tail splitting and Barbalat (Proposition 2.6).

Proposition 2.2 (Pair bound, `pair_bound`). *For any $\alpha_j, \alpha_k, \alpha_j^*, \alpha_k^* \in (0, 1)$:*

$$\alpha_j^*(\alpha_k - \alpha_k^*)^2 \left(\alpha_k + \frac{1}{\alpha_k^*} \right) + \alpha_k^*(\alpha_j - \alpha_j^*)^2 \left(\alpha_j + \frac{1}{\alpha_j^*} \right) \geq (\alpha_j - \alpha_j^*)(\alpha_k - \alpha_k^*)(2 - \alpha_j^2 - \alpha_k^2). \quad (6)$$

Proof. The inequality follows from an algebraic decomposition plus a sign case split. With $p_j = \alpha_j - \alpha_j^*$ and $p_k = \alpha_k - \alpha_k^*$, clearing denominators by $\alpha_j^* \alpha_k^* > 0$ yields:

$$\alpha_j^* \alpha_k^* \cdot (\text{LHS} - \text{RHS}) = (\alpha_j^* p_k - \alpha_k^* p_j)^2 + (\alpha_j^*)^2 \alpha_k \alpha_k^* p_k^2 + (\alpha_k^*)^2 \alpha_j \alpha_j^* p_j^2 + \alpha_j^* \alpha_k^* (\alpha_j^2 + \alpha_k^2) p_j p_k.$$

When $p_j p_k \leq 0$: the left side of (6) is non-negative (each summand has form (positive) $\cdot p^2 \cdot$ (positive) ≥ 0), while the right side $p_j p_k (2 - \alpha_j^2 - \alpha_k^2) \leq 0$ since $p_j p_k \leq 0$ and $2 - \alpha_j^2 - \alpha_k^2 > 0$ for $\alpha_j, \alpha_k \in (0, 1)$. Therefore $\text{LHS} \geq 0 \geq \text{RHS}$. When $p_j p_k > 0$: the algebraic decomposition plus sign case split gives a positive lower bound, so the expression is strictly positive and hence non-negative. This is verified algebraically in `pair_bound_clear_denom`. $\square$

Proposition 2.3 (Continuum pair bound, `continuum_pair_nonneg`). *The double integral*

$$\int_{\mathbb{R}} \int_{\mathbb{R}} \left[\alpha^*(\omega_1) (\alpha(\omega_2) - \alpha^*(\omega_2))^2 \left(\alpha(\omega_2) + \frac{1}{\alpha^*(\omega_2)} \right) + \alpha^*(\omega_2) (\alpha(\omega_1) - \alpha^*(\omega_1))^2 \left(\alpha(\omega_1) + \frac{1}{\alpha^*(\omega_1)} \right) \right. \\ \left. - (\alpha(\omega_1) - \alpha^*(\omega_1)) (\alpha(\omega_2) - \alpha^*(\omega_2)) (2 - \alpha(\omega_1)^2 - \alpha(\omega_2)^2) \right] g(\omega_1) g(\omega_2) d\omega_1 d\omega_2 \geq 0.$$

Proof. Mathlib's `integral_nonneg` applied twice (once in ω_1 , once in ω_2) reduces this to the pointwise inequality of Proposition 2.2. $\square$

Proposition 2.4 (Derived monotonicity from the pair bound, `hV_rate`). *Under the hypotheses of Theorem 1.2, $V'(t) \leq 0$ for all $t > 0$. Convergence $V \rightarrow 0$ follows from body persistence combined with Barbalat's lemma via the body-tail split: on $\{\gamma \leq M\}$, body persistence gives a coercive lower bound on $-V'$; on $\{\gamma > M\}$, the tail contribution vanishes as $M \rightarrow \infty$ since $\int \gamma g < \infty$. This is proved from the pair bound and persistence; it is not assumed.*

Proof sketch. Substituting the OA equation (2) into the Leibniz formula:

$$V'(t) = \int_{\mathbb{R}} 2(\alpha - \alpha^*) \left[-\gamma\alpha + \frac{K}{2}r(1 - \alpha^2) \right] g \, d\omega.$$

Using the equilibrium condition (4) to write $\gamma\alpha^* = (K/2)r^*(1 - (\alpha^*)^2)$, one computes:

$$2(\alpha - \alpha^*)\dot{\alpha} = K(\alpha - \alpha^*) \left[r(1 - \alpha^2) - r^*(1 - (\alpha^*)^2) - \frac{2\gamma}{K}(\alpha - \alpha^*) \right].$$

Expanding and grouping, the integrand splits into:

- a term proportional to $(r - r^*) \cdot D$ where $D = \int (\alpha - \alpha^*)g \, d\omega = r - r^*$, giving $K(r - r^*)^2 \leq KV$ (by Cauchy-Schwarz);
- a term $-Kr^* \cdot Q$ where $Q = \int (\alpha - \alpha^*)^2(\alpha + 1/\alpha^*)g \, d\omega$;
- a term $K \cdot DS$ where $S = \int (\alpha - \alpha^*)(1 - \alpha^2)g \, d\omega$.

The continuum pair bound (Proposition 2.3) gives $DS \leq r^*Q$, so $K \cdot DS - Kr^*Q \leq 0$, and the pair bound already yields $V'(t) \leq 0$. To obtain convergence $V \rightarrow 0$, one uses the body-tail split: on the body $\{\gamma \leq M\}$, body persistence gives $\alpha(\omega, t) \geq \delta(M) > 0$, which provides a coercive lower bound $Q_{\text{body}} \geq \delta(M) \cdot \delta^* \cdot V_{\text{body}}$; the tail $\{\gamma > M\}$ contributes at most $\int_{\gamma > M} \gamma g \, d\omega \rightarrow 0$ as $M \rightarrow \infty$. Barbalat's lemma then gives $V \rightarrow 0$. The full computation is verified in `ContinuumUniformRate.lean`. $\square$

2.3 Step 3: V is antitone

Proposition 2.5 (V antitone, `lyapunov_antitone`). $V(t)$ is non-increasing on $[0, \infty)$.

Proof. By Proposition 2.4, $V'(t) \leq 0$ for all $t > 0$. Since V is continuous (Proposition 2.1), Mathlib's `antitoneOn_of_deriv_nonpos` applies directly on $[0, \infty)$. $\square$

2.4 Step 4: Body-tail split and Barbalat

Proposition 2.6 (Body-tail convergence, `supercritical_coercive_drops`). For any $\varepsilon > 0$, there exists $T > 0$ such that $V(t) < \varepsilon$ for all $t > T$.

Proof. **Step A (Tail control).** Decompose

$$V = V_{\text{body},M} + V_{\text{tail},M}, \quad V_{\text{body},M} := \int_{\gamma \leq M} (\alpha - \alpha^*)^2 g, \quad V_{\text{tail},M} := \int_{\gamma > M} (\alpha - \alpha^*)^2 g.$$

Since $|\alpha - \alpha^*| \leq 1$ and $(\alpha - \alpha^*)^2 \leq 1$:

$$V_{\text{tail},M} \leq \int_{\gamma > M} g \, d\omega.$$

Since $\int \gamma g < \infty$, Markov's inequality gives $\int_{\gamma > M} g \leq \frac{1}{M} \int \gamma g \rightarrow 0$ as $M \rightarrow \infty$. Thus for every $\varepsilon > 0$, there exists M_0 such that $V_{\text{tail},M_0}(t) < \varepsilon/2$ for all t .

Step B (Body convergence). Fix $M = M_0$. On $\{\gamma \leq M\}$, body persistence (derived in Step 3 of the proof of Theorem 1.2) gives $\alpha(\omega, t) \geq \delta(M) > 0$. The body-restricted pair coercivity then gives a positive lower bound:

$$-V'(t) \geq K \cdot \delta(M) \cdot \delta^*(M) \cdot \mu(\gamma \leq M) \cdot V_{\text{body},M}(t),$$

where $\delta^*(M) = \min_{\gamma \leq M} \alpha^*(\omega) > 0$. Since V is antitone and $V' \leq -c \cdot V_{\text{body},M}$ with $c > 0$: the time-integral $\int_0^\infty V_{\text{body},M}(t) dt$ converges (bounded by $V(0)/c$). The derivative of $V_{\text{body},M}$ satisfies

$$\left| \frac{d}{dt} V_{\text{body},M}(t) \right| = \left| \int_{\gamma \leq M} 2(\alpha - \alpha^*) \dot{\alpha} g \right| \leq 2(M + K) \mu(\gamma \leq M),$$

since $|\alpha - \alpha^*| \leq 1$ and $|\dot{\alpha}| \leq M + K$ on $\{\gamma \leq M\}$. A bounded derivative gives Lipschitz, hence uniform continuity of $V_{\text{body},M}$. Since $V_{\text{body},M} \geq 0$ and $\int_0^\infty V_{\text{body},M}(t) dt < \infty$, Barbalat's lemma gives $V_{\text{body},M}(t) \rightarrow 0$.

Step C (Combine). For t large enough: $V_{\text{body},M_0}(t) < \varepsilon/2$ (Step B) and $V_{\text{tail},M_0}(t) < \varepsilon/2$ (Step A). Therefore $V(t) < \varepsilon$.

This avoids any appeal to uniform continuity of V' . The Lean formalization uses `coercive_drop_from_persistence` in `ContinuumBarbalat.lean`, which takes: non-negativity, antitonicity, body persistence, and tail vanishing. $\square$

Remark 2.7. In the n -pole case (Section 5), the stronger exponential rate $V(t+1) \leq e^{-\mu} V(t)$ with $\mu = K c_{\min} \delta \delta^*$ holds via Gronwall comparison (`comparison_decay_interval` in `GronwallBridge.lean`).

2.5 Step 5: $V \rightarrow 0$

Proposition 2.8 ($V \rightarrow 0$, `coercive_drop_from_persistence`). $V(t) \rightarrow 0$ as $t \rightarrow +\infty$.

Proof. By Proposition 2.6, $V(t) \rightarrow 0$ via the body-tail split and Barbalat's lemma. Since V is antitone (Proposition 2.5) and bounded below by 0, the limit exists, and the body-tail argument forces it to be zero. The formalization uses `coercive_drop_from_persistence` in `ContinuumBarbalat.lean`, which takes: non-negativity, antitonicity, body persistence, and tail vanishing. $\square$

2.6 Step 6: $r \rightarrow r^*$ via Cauchy-Schwarz

Proposition 2.9 ($r \rightarrow r^*$, closing step). $V(t) \rightarrow 0$ implies $r(t) \rightarrow r^*$.

Proof. By the self-consistency relation and the equilibrium formula:

$$r(t) - r^* = \int_{\mathbb{R}} (\alpha(\omega, t) - \alpha^*(\omega)) g(\omega) d\omega.$$

By the Cauchy-Schwarz inequality (or Jensen applied to the square function):

$$(r(t) - r^*)^2 = \left(\int_{\mathbb{R}} (\alpha - \alpha^*) g d\omega \right)^2 \leq \int_{\mathbb{R}} (\alpha - \alpha^*)^2 g d\omega = V(t).$$

This is formalized as `sq_integral_le_integral_sq` in the Lean code. Since $V(t) \rightarrow 0$, we have $(r(t) - r^*)^2 \rightarrow 0$, hence $r(t) \rightarrow r^*$ in the metric sense. $\square$

Proof of Theorem 1.2. Steps 1 through 6 above, composed in order, constitute the complete proof. In Lean:

- (1) `leibniz_oa_lyapunov` gives continuity and differentiability of V .
- (2) The pair bound (`continuum_pair_nonneg`) and Leibniz together give $V' \leq 0$.
- (3) `lyapunov_antitone` gives that V is non-increasing.
- (4) Cauchy-Schwarz gives $r(t) \geq r^* - \sqrt{V(0)} > 0$ (order parameter persistence).
- (5) ODE scalar barrier gives body persistence on $\{\gamma \leq M\}$ for each $M > 0$.

(6) `coercive_drop_from_persistence` gives $V \rightarrow 0$ via body-tail split + Barbalat.

(7) `sq_integral_le_integral_sq` gives $(r - r^*)^2 \leq V$, so $r \rightarrow r^*$.

The end-to-end theorem is `real_scalar_complete` in `RealScalarComplete.lean`, which wires the chain and derives body persistence internally from the basin condition. $\square$

2.7 Step 7: Pair rigidity (characterization of $V' = 0$)

Although not needed for the convergence proof itself, the pair bound also characterizes when $V' = 0$.

Proposition 2.10 (Pair rigidity, `pair_eq_zero_iff`). *The pair integrand (6) equals zero if and only if $\alpha_j = \alpha_j^*$ and $\alpha_k = \alpha_k^*$.*

Proof. The algebraic decomposition plus sign case split in the proof of Proposition 2.2 shows:

$$\alpha_j^* \alpha_k^* \cdot (\text{LHS} - \text{RHS}) = (\alpha_j^* p_k - \alpha_k^* p_j)^2 + (\alpha_j^*)^2 \alpha_k^* p_k^2 + (\alpha_k^*)^2 \alpha_j^* p_j^2 + \alpha_j^* \alpha_k^* (\alpha_j^2 + \alpha_k^2) p_j p_k.$$

When $p_j p_k > 0$: the second and third terms are strictly positive, so the sum is positive and the integrand cannot be zero unless $p_j = p_k = 0$. When $p_j p_k \leq 0$: each of the three original terms of (6) is non-negative (since $\alpha_j + 1/\alpha_j^* > 0$ and $2 - \alpha_j^2 - \alpha_k^2 > 0$), so all must individually vanish, giving $p_j = p_k = 0$. This is formalized as `pair_eq_zero_iff` in `ContinuumLyapunov.lean`. $\square$

Remark 2.11 (LaSalle structure). *Proposition 2.10 shows that $\{V' = 0\} = \{\alpha = \alpha^*\}$. This is the LaSalle invariance principle characterization and is used in the alternative Path B proof (`ContinuumGlobalStability.lean`) where scalar asymptotic autonomy of each oscillator gives pointwise convergence and dominated convergence gives $V \rightarrow 0$.*

3 The Lorentzian case: fully end-to-end

For the Lorentzian distribution $g(\omega) = \gamma/(\pi(\omega^2 + \gamma^2))$, the OA equation reduces to the scalar Bernoulli ODE

$$\dot{r} = \left(\frac{K}{2} - \gamma\right) r - \frac{K}{2} r^3. \quad (7)$$

The equilibrium is $r^* = \sqrt{1 - 2\gamma/K}$ for $K > 2\gamma$.

Theorem 3.1 (Lorentzian convergence, `lorentzian_ode_global_stability_complete`). *For $K > 2\gamma > 0$ and any $r_0 \in (0, 1)$, the unique solution of (7) satisfies $r(t) \rightarrow r^*$.*

The proof chain in Lean is:

$$(K, \gamma, r_0) \xrightarrow{\text{Bernoulli}} \text{LorentzianContinuousSolution} \rightarrow \text{LorentzianSolution} \rightarrow \text{KuramotoData} \rightarrow r(t)$$

3.1 Proof: Bernoulli formula

Proposition 3.2 (Existence via Bernoulli, `LorentzianExistence`). *The ODE (7) has an explicit solution $r(t) = (w(t))^{-1/2}$ where $w(t) = w_0 e^{-(K-2\gamma)t} + (K/(K-2\gamma))(1 - e^{-(K-2\gamma)t})$ and $w_0 = r_0^{-2}$. This function satisfies `HasDerivAt` at every $t \geq 0$, verified by differentiating the explicit formula.*

3.2 Proof: Monotonicity

Proposition 3.3 (Monotonicity, `LorentzianFromODE`). *If $r_0 < r^*$ then r is non-decreasing. If $r_0 > r^*$ then r is non-increasing. In both cases $r \in (0, 1)$ for all t .*

Proof. From (7), $\dot{r} = (K/2)r(1 - 2\gamma/K - r^2) = (K/2)r((r^*)^2 - r^2)$. So $\dot{r} > 0$ when $r < r^*$ and $\dot{r} < 0$ when $r > r^*$. Invariance $r \in (0, 1)$ follows from the barrier structure at $r = 0$ and $r = 1$. $\square$

3.3 Proof: Convergence

Proof of Theorem 3.1. r is monotone and bounded, so $r(t) \rightarrow L$ for some $L \in [0, 1]$. Taking $t \rightarrow \infty$ in (7) (using continuity of the right side and $\dot{r} \rightarrow 0$ from the monotone convergence theorem): $0 = (K/2 - \gamma)L - (K/2)L^3$. The nonzero solutions are $L = \pm r^*$; since $r > 0$ always, $L = r^*$. $\square$

The complete proof is assembled in `lorentzian_ode_global_stability_complete` with zero additional axioms beyond Lean’s standard foundations, verified in `LorentzianInstance.lean`.

4 Lean 4 verification

4.1 Overall status

The formalization spans **255 Lean 4 files** (commit 58b6194, Mathlib v4.30.0-rc1). The declaration `real_scalar_complete` depends on no project axioms and no `sorry`; the imported development contains unrelated `sorry`s in theorems outside this dependency chain. The formalization uses only Lean 4’s standard foundations.

There are two tiers of completeness:

Tier 1 — Lorentzian (fully end-to-end). The theorem `lorentzian_ode_global_stability_complete` takes only the physical parameters (K, γ, r_0) and constructs the complete proof: solution existence via the explicit Bernoulli formula, all solution properties, and convergence $r \rightarrow r^*$. Zero axioms, zero assumed fields.

Tier 2 — Finite first moment (fully self-contained). The theorem `kuramoto_stability` in `KuramotoFinal.lean` derives everything internally:

1. Leibniz rule $\rightarrow V$ differentiable (finite first moment provides the integrable dominator $2(\gamma(\omega) + K)$)
2. Pair bound $\rightarrow \dot{V} \leq 0$
3. V antitone $\rightarrow V(t) \leq V(0) < (r^*)^2$
4. Cauchy–Schwarz $\rightarrow r(t) \geq r^* - \sqrt{V(0)} > 0$ (order parameter persistence)
5. ODE scalar barrier $\rightarrow$ body persistence on $\{\gamma \leq M\}$ for each $M > 0$
6. Body-tail split + Barbalat $\rightarrow V \rightarrow 0 \rightarrow r(t) \rightarrow r^*$

No persistence hypothesis is assumed externally. No uniform-in- ω lower bound is assumed or derived. The only initial conditions are $V(0) < (r^*)^2$ and body coherence at $t = 0$. **0 sorry, 0 axioms.**

Tier 3 — N-pole (any finite n). The theorem `kuramoto_solved` provides an alternative proof path for discrete approximations with explicit exponential rate $\dot{V} \leq -\mu V$ where $\mu = K c_{\min} \delta \delta^*$.

4.2 The proof chain in Lean

The following table shows how each proposition in Section 2 corresponds to a Lean theorem:

Proposition	Lean name	File
Leibniz rule (Prop. 2.1)	leibniz_oa_lyapunov	GeneralGMainTheorem.lean
Pair bound (Prop. 2.2)	pair_bound	L2Lyapunov.lean
Continuum pair (Prop. 2.3)	continuum_pair_nonneg	ContinuumLyapunov.lean
Monotonicity bound (Prop. 2.4)	hV_rate	ContinuumUniformRate.lean
V antitone (Prop. 2.5)	lyapunov_antitone	GeneralGMainTheorem.lean
Exp. drop (Prop. 2.6)	supercritical_coercive_drops	GeneralGMainTheorem.lean
$V \rightarrow 0$ (Prop. 2.8)	coercive_drop_from_persistence	ContinuumBarbalat.lean
$r \rightarrow r^*$ (Prop. 2.9)	sq_integral_le_integral_sq	GeneralGMainTheorem.lean
Pair rigidity (Prop. 2.10)	pair_eq_zero_iff	ContinuumLyapunov.lean

Key Mathlib lemmas used in the proof chain:

- `hasDerivAt_integral_of_dominated_loc_of_deriv_le` — Leibniz differentiation under the integral sign;
- `antitoneOn_of_deriv_nonpos` — monotone convergence from a derivative bound;
- `integral_nonneg` — applied pointwise to transfer the pair bound to integrals;
- `MeasureTheory.continuous_of_dominated` — dominated convergence for continuity of V .

4.3 Nine independent proof paths

The formalization contains nine complementary, sorry-free proof paths of basin stability or closely related statements. Each addresses a different mathematical angle:

File	Proof strategy
GeneralGMainTheorem	Leibniz + pair bound + coercive Barbalat (Sections 2.1–2.6; this paper)
MainTheorem	Gap exclusion + Lipschitz trapping (discrete, self-consistency)
ContinuousStability	IVT trapping (continuous-time)
NPoleGlobalStability	L^2 Barbalat + persistence drops (n -pole systems)
GronwallBridge + NPoleInstance	L^2 Gronwall + exponential rate (continuous n -pole)
ContinuumLyapunov	$\dot{V} \leq 0$ directly (pair bound $\rightarrow$ integrals)
AntitoneConvergence	$V \rightarrow L \geq 0$ (bounded antitone sequences)
ContinuumGlobalStability	Scalar asymptotic autonomy (Path B: each $\alpha(\omega, \cdot) \rightarrow \alpha^*(\omega)$)
InstabilityExclusion	V antitone + instability escape $\Rightarrow V \rightarrow 0$ (no persistence needed)

4.4 Additional formalized results

Beyond global stability, the formalization establishes:

- **Complete trifurcation:** $K < K_c \Rightarrow r(t) \rightarrow 0$; $K = K_c \Rightarrow r(t) \rightarrow 0$ at rate $O(1/\sqrt{t})$; $K > K_c \Rightarrow r(t) \rightarrow r^*$.
- **Bifurcation monotonicity:** $r^*(K)$ is increasing in K for $K > K_c$.
- **Instability of incoherence:** proved via Chetaev’s theorem for $K > K_c$.

- **Lorentzian trifurcation:** $K < 2\gamma \Rightarrow r \rightarrow 0$, $K = 2\gamma \Rightarrow r \rightarrow 0$, $K > 2\gamma \Rightarrow r \rightarrow r^*$.

To verify: `cd kuramoto-lean && lake build KuramotoLean.RealScalarComplete` (commit 58b6194).

5 The L^2 Lyapunov approach for n -pole systems

This section describes the cleanest of the nine proof paths for finite approximations: an explicit Lyapunov function for n -pole systems that decreases monotonically and decays exponentially. The argument is a finite-dimensional specialization of the continuum proof in Section 2.

5.1 Setup

An n -pole system is a finite approximation to the continuum Kuramoto model in which the frequency distribution g is replaced by a discrete measure $\sum_{k=1}^n c_k \delta_{\omega_k}$. The state is $(\alpha_k)_{k=1}^n \in (0, 1)^n$ and the order parameter is $r = \sum_k c_k \alpha_k$.

Let α_k^* denote the PLS values. Define the L^2 Lyapunov function:

$$V(t) = \sum_{k=1}^n c_k |\alpha_k(t) - \alpha_k^*|^2. \quad (8)$$

5.2 Main decay estimate

Theorem 5.1 (L^2 Lyapunov decay, `12.exponential.rate`). *For any n -pole Kuramoto system with $K > K_c$, there exist constants $c_{\min} > 0$, $\delta > 0$, $\delta^* > 0$ depending only on K , g , and the pole weights such that*

$$\frac{dV}{dt} \leq -K c_{\min} \delta \delta^* V.$$

In particular, $V(t) \leq V(0) e^{-K c_{\min} \delta \delta^* t} \rightarrow 0$ exponentially.

Proof outline. Differentiate (8) using the Kuramoto ODEs. Each pair (α_k, α_k^*) contributes a term involving $\text{Re}((\alpha_k - \alpha_k^*)(\dot{\alpha}_k - \dot{\alpha}_k^*))$. Writing $\dot{\alpha}_k - \dot{\alpha}_k^*$ in terms of $r - r^*$ and using the finite sum version of the pair bound (Proposition 2.2), together with the uniform lower bound $c_{\min} = \min_k c_k$:

$$\frac{dV}{dt} \leq -K c_{\min} \delta \delta^* V.$$

This is formalized in `UniformRate.lean` and `L2Lyapunov.lean` (0 sorry). $\square$

Remark 5.2 (Uniformity in n). *The bound $dV/dt \leq -K c_{\min} \delta \delta^* V$ is uniform in n : the constant does not depend on the number of poles, only on the weight floor $c_{\min}$ and the gap parameters δ , δ^* . In the continuum limit (Proposition 2.4), the explicit exponential rate is replaced by the body-tail split and Barbalat's lemma, since $c_{\min} \rightarrow 0$ as the discrete measure approximates a continuous density.*

5.3 Convergence from the Lyapunov bound

Global stability follows immediately: $V(t) \rightarrow 0$ implies $\|\alpha(t) - \alpha^*\|_{L^2(g)}^2 \rightarrow 0$, which in particular forces $|r(t) - r^*| \rightarrow 0$ by Cauchy-Schwarz (Proposition 2.9). The chain is:

$$V(0) < \infty \xrightarrow{\text{Thm. 5.1}} V(t) \rightarrow 0 \xrightarrow{\text{Cauchy-Schwarz}} |r(t) - r^*| \rightarrow 0.$$

This proof path (`NPoleGlobalStability.lean`, `GronwallBridge.lean`, `NPoleInstance.lean`) requires no Barbalat lemma beyond the monotone convergence theorem, and no analyticity of g —only the existence of the PLS and the gap bound.

Remark 5.3 (Local vs. global stability). *For n -pole systems (any finite n), the theorem `kuramoto_solved` gives global stability: any $r(0) > 0$ converges. Persistence is automatic in finite dimensions. For the continuum, the theorem `kuramoto_stability` gives stability within the basin $V(0) < (r^*)^2$. Removing this restriction requires proving that $r(t)$ stays uniformly positive for arbitrary initial data—equivalent to proving instability of the incoherent state $\alpha = 0$ for the nonlinear continuum OA system. This remains an open problem: no existing paper proves it for general g . A natural approach is passage to the limit from n -pole to continuum via continuous dependence of OA solutions on the measure g .*

6 Complex OA: rotation cancellation and conditional stability

The pair bound argument of Sections 2–2.6 applies directly to the scalar model (2). We now show it extends to the standard complex OA equation (1).

6.1 Dietert energy identity

Proposition 6.1 (`psi_deriv_eq_K_eta_sq`, 0 sorry). *Define the Dietert energy $\Psi(t) := -\int \log(1 - |z(\omega, t)|^2) g \, d\omega$. Then*

$$\frac{d\Psi}{dt} = K|\eta(t)|^2 \geq 0.$$

Proof. By chain rule, $\frac{d}{dt}[-\log(1 - |z|^2)] = \frac{d|z|^2/dt}{1 - |z|^2}$. From the complex OA, $d|z|^2/dt = K \operatorname{Re}(\eta z)(1 - |z|^2)$ (`complexOa_normSq_deriv`, 0 sorry). Hence the pointwise integrand is $K \operatorname{Re}(\eta z)$. Integrating: $d\Psi/dt = K \int \operatorname{Re}(\eta z) g = K \operatorname{Re}(\eta \bar{\eta}) = K|\eta|^2$, using $\bar{\eta} = \int z g$ for real-valued g (`complex_eta_integral_identity`, 0 sorry). $\square$

6.2 Rotation cancellation

The L^2 Lyapunov function $V(t) = \int |z - z^*|^2 g \, d\omega$ has derivative

$$V'(t) = 2 \int \operatorname{Re}(\overline{(z - z^*)} \cdot \dot{z}) g \, d\omega.$$

Subtracting the equilibrium equation $0 = -i\omega z^* + (K/2)(\bar{\eta}^* - \eta^* z^{*2})$:

$$\dot{z} - 0 = -i\omega(z - z^*) + \frac{K}{2}[(\eta - \eta^*)(1 - z^2) + \eta^*(z^{*2} - z^2)].$$

Proposition 6.2 (Rotation cancellation, `rotation_zero_in_error`, 0 sorry). $\operatorname{Re}(\overline{(z - z^*)} \cdot (-i\omega(z - z^*))) = \operatorname{Re}(-i\omega|z - z^*|^2) = 0$.

Proof. $|z - z^*|^2 \in \mathbb{R}$, so $-i\omega|z - z^*|^2$ is purely imaginary. $\square$

The rotation cancellation removes the rotation contribution from V' , but the remaining nonlinear coupling terms differ structurally from the scalar case: the coupling involves $\eta \in \mathbb{C}$ (not $r \in \mathbb{R}$), and the quadratic terms $\eta^*(z^{*2} - z^2)$ do not factor as cleanly as in the scalar pair bound. The convergence $V(t) \rightarrow 0$ is therefore established via nonlinear Landau damping [9]: for Sobolev-regular perturbations of the PLS, phase mixing of drifting oscillators gives algebraic decay $|V(t)| = O(t^{-s})$, valid universally for all $K > K_c$ (stated as axiom `landau_damping_kuramoto`). The complex OA stability result relies on this external Landau damping input (cited as axiom).

6.3 Symmetry preservation

Proposition 6.3 (`complex_0a_symmetry_preserved`, 0 sorry). *For symmetric g with symmetric initial data, $z(\omega, t) = \overline{z(-\omega, t)}$ for all t . Consequently, $\eta(t) \in \mathbb{R}$.*

Proof. Define $w(\omega, t) := \overline{z(-\omega, t)}$. Then w satisfies the same *coupled* OA system: conjugating the ODE and using $g(-\omega) = \overline{g(\omega)}$ gives $\dot{w}(\omega, t) = -i\omega w + (K/2)(\bar{\eta}_w - \eta_w w^2)$ where $\eta_w = \int \bar{w} g = \int z(-\cdot) g = \int z g$ (by measure-preserving negation) $= \bar{\eta}_z$ (by `integral_conj`). The algebraic identity `complex0aRHS_conj_neg` gives $f(-\omega, \eta, z) = \overline{f(\omega, \bar{\eta}, \bar{z})}$. Since $w(\omega, 0) = z(\omega, 0)$ and both satisfy the same coupled system, *coupled* ODE uniqueness gives $w = z$, establishing symmetry and $\eta \in \mathbb{R}$ simultaneously without circularity (0 sorry). $\square$

6.4 Full Kuramoto PDE stability

Theorem 6.4 (Conditional bridge lemma, `full_kuramoto_pde_stability_via_axiom`, 0 sorry + 1 axiom). *For the full Kuramoto–Sakaguchi PDE with analytic symmetric g , $K > K_c$: if the OA order parameter converges ($\|\eta_{\text{OA}}(t)\| \rightarrow r^*$) and the full PDE approximates the OA trajectory ($\|\eta_{\text{full}} - \eta_{\text{OA}}\| \rightarrow 0$), then $\|\eta_{\text{full}}(t)\| \rightarrow r^*$. This theorem is conditional: both hypotheses are cited axioms, not machine-checked.*

Proof. Triangle inequality: $|\|\eta_{\text{full}}\| - \|\eta_{\text{OA}}\|| \leq \|\eta_{\text{full}} - \eta_{\text{OA}}\| \rightarrow 0$. Squeeze gives convergence (`squeeze_zero_norm + abs_norm_sub_norm_le`, 0 sorry). $\square$

Remark 6.5 (Hypotheses of Theorem 6.4). *The hypothesis $\|\eta_{\text{full}} - \eta_{\text{OA}}\| \rightarrow 0$ follows from OA manifold exponential attractivity [2] (axiom). The hypothesis $\|\eta_{\text{OA}}\| \rightarrow r^*$ follows from nonlinear Landau damping [9] (axiom) giving $V \rightarrow 0$, then Cauchy–Schwarz (machine-checked). The full chain is `kuramoto_stability_universal` (0 sorry, 2 axioms).*

7 Discussion

7.1 Four tiers of formalization

The results are organized into four tiers of decreasing certainty:

Theorem A: Real scalar complete (`real_scalar_complete`). Fully formalized, 0 sorry, 0 axioms. Basin stability for the scalar dissipative model (2) with finite first moment $\int \gamma g < \infty$. When $\gamma(\omega) \lesssim 1 + |\omega|$: Gaussian, Student- t ($\nu > 1$), compact support, mixtures.

Theorem B: Lorentzian (`lorentzian_ode_global_stability_complete`). Fully formalized, 0 sorry, 0 axioms. Global stability for the Bernoulli ODE with explicit solution formula. Finite-dimensional, no measure theory needed.

Theorem C: n -pole (`kuramoto_solved`). Fully formalized, 0 sorry, 0 axioms. Finite-dimensional, exponential decay rate $\mu = K c_{\min} \delta \delta^*$. Global stability for any $r(0) > 0$.

Theorem D: Complex OA + full PDE (`kuramoto_stability_universal`). Conditional, 0 sorry, 2 axioms. The complex OA convergence relies on nonlinear Landau damping [9] (axiom). The PDE-to-OA bridge relies on OA manifold attractivity [2] (axiom). Both axioms have opaque (non-vacuous) conditions encoded as opaque Prop types.

7.2 Theorem table

Name	Model	Hypotheses	Conclusion	Sor
<code>real_scalar_complete</code>	Scalar dissipative	$\int \gamma g < \infty$, $V(0) < (r^*)^2$	$r(t) \rightarrow r^*$	0
<code>lorentzian_ode_global_stability_complete</code>	Lorentzian ODE	$K > 2\gamma$, $r_0 \in (0, 1)$	$r(t) \rightarrow r^*$	0
<code>kuramoto_solved</code>	n -pole	$K > K_c$, $r(0) > 0$	$r(t) \rightarrow r^*$	0
<code>kuramoto_stability_universal</code>	Complex OA + PDE	Analytic g , $K > K_c$	$\ \eta\ \rightarrow r^*$	0

7.3 Novelty

This work contributes four things that are new:

1. **First machine-checked Kuramoto result.** To our knowledge, this is the first formalization in any proof assistant (Lean, Coq, Isabelle) addressing any aspect of the Kuramoto model.
2. **Novel proof technique.** The pair bound (Proposition 2.2)—a pointwise algebraic inequality lifted to the continuum double integral via Fubini—has no precedent in the Kuramoto literature. Prior approaches used Fourier analysis [1], Volterra equations [2], or spectral theory [6].
3. **Rotation cancellation.** The rotation term $-i\omega$ in the complex OA cancels in the V derivative after equilibrium subtraction (Proposition 6.2). This removes the rotation contribution from V' , but the remaining nonlinear coupling terms differ structurally from the scalar case. The complex OA convergence relies on external Landau damping input (cited as axiom).
4. **Finite first moment coverage.** Prior rigorous OA analysis was limited to Lorentzian/rational distributions. The finite first moment condition $\int \gamma g d\omega < \infty$ covers Student- t with $\nu > 1$ (including the infinite-variance case $1 < \nu \leq 2$), Gaussian, compact support, and all mixtures.

7.4 Limitations

1. **Basin of attraction.** The continuum result (Theorem A) requires $V(0) < (r^*)^2$, an explicit but *local* condition. True global stability (any $r(0) > 0$) is proved only for Lorentzian (Theorem B) and n -pole (Theorem C) systems.
2. **Two cited axioms with opaque conditions.** Theorem D relies on two published results stated as Lean axioms with opaque (non-vacuous) conditions: (a) nonlinear Landau damping [9] for $V \rightarrow 0$, requiring linear stability, Sobolev smallness, and regularity of g ; and (b) OA manifold attractivity [2] for the PDE-to-OA bridge. The conditions are encoded as `opaque Prop` types that cannot be trivially discharged. Formalizing the underlying spectral theory remains future work.
3. **Conclusion is $\text{Re}(\eta)^2$, not $|\eta|^2$.** On the symmetric subspace $\eta \in \mathbb{R}$ (proved, 0 sorry), so $\text{Re}(\eta)^2 = |\eta|^2$. But the Lean type states $\text{Re}(\eta)^2 \rightarrow (r^*)^2$ rather than $|\eta|^2 \rightarrow (r^*)^2$.
4. **Scalar model scope.** The scalar dissipative model (2) is a separate dynamical system from the standard complex OA (1).

7.5 Open problems

1. Remove the $V(0) < (r^*)^2$ basin condition (global stability for any $r(0) > 0$).
2. Formalize the nonlinear Landau damping axiom (currently citing [9]).
3. Formalize the OA manifold attractivity axiom (currently citing [2]).
4. Extend to distributions without finite first moment (Student- t with $\nu \leq 1$) beyond the Lorentzian special case.

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